

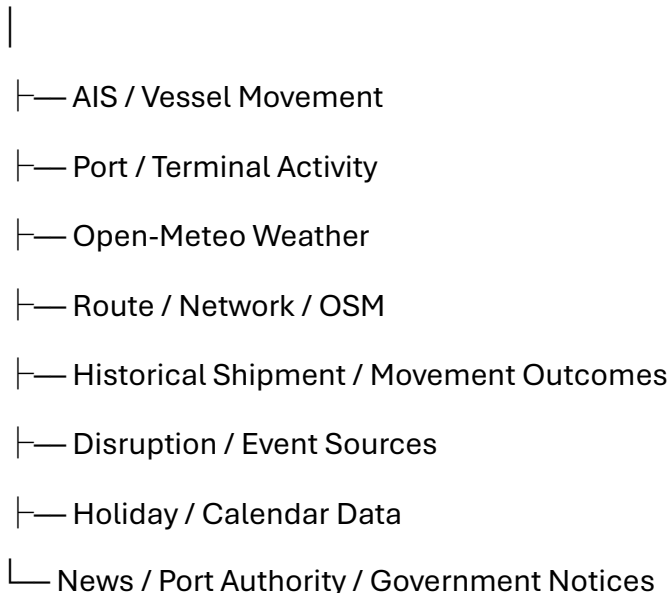
LOGIX — Complete Data Column Specification

Purpose: This document is the canonical data contract for implementing LOGIX. OpenCode should use these schemas as the target structure for ingestion, normalization, feature engineering, ML training, graph propagation, Monte Carlo simulation, and dashboard outputs.

Important: Not every external data source will provide every column. The implementation must **never fabricate unavailable source data**. Columns marked derived should be calculated from real source data; columns marked optional should be nullable when the source does not provide them.

1. Data Architecture

External Sources



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RAW PARQUET

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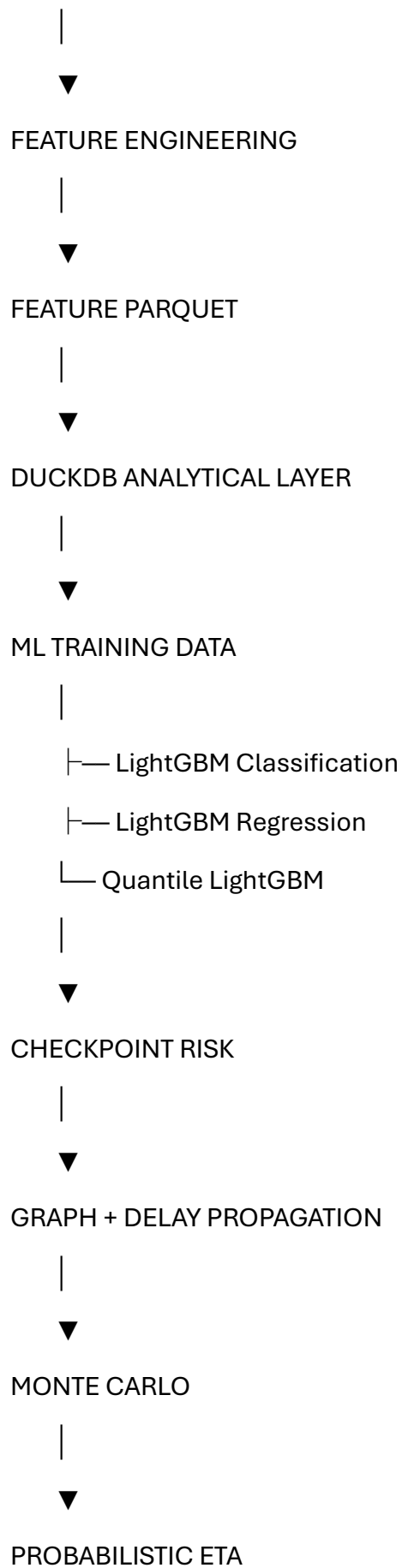


DATA QUALITY + VALIDATION

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NORMALIZED PARQUET



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DEADLINE RISK

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WHAT-IF / INTERVENTION / ROUTE DECISION

2. Universal Columns

Every normalized dataset should preserve provenance wherever possible.

These columns are recommended across the normalized data layer.

Column	Type	Required	Description
record_id	string	Yes	Unique row/observation ID
source	string	Yes	Data source/provider
source_record_id	string	No	Original source record ID
source_url	string	No	Source/API URL
ingested_at	timestamp	Yes	When LOGIX ingested the record
source_updated_at	timestamp	No	Source update timestamp
data_version	string	No	Dataset/source version
quality_status	string	Yes	valid, flagged, invalid
quality_flags	string/list	No	Validation issues
is_duplicate	boolean	Yes	Duplicate indicator

3. Shipment / Delivery Data

File:

data/raw/shipments/

data/silver/shipments/

This represents the business-level shipment/request.

Column	Type	Required	Description
shipment_id	string	Yes	Unique shipment identifier
route_id	string	Yes	Route assigned to shipment
origin_node_id	string	Yes	Origin node
destination_node_id	string	Yes	Destination node
departure_time	timestamp	Yes	Actual/planned departure depending on record
planned_departure_time	timestamp	No	Planned departure
actual_departure_time	timestamp	No	Actual departure
required_delivery_time	timestamp	Yes	Customer-required delivery deadline
transport_mode	string	Yes	Sea/road/rail/air/multimodal
cargo_type	string	No	Cargo category
cargo_priority	string	No	Priority category
service_level	string	No	Service-level requirement
cargo_value	float	No	Cargo value
penalty_cost	float	No	Cost associated with missing deadline
customer_priority	string	No	Customer importance
planned_transit_hours	float	No	Planned transit duration
actual_transit_hours	float	No	Actual transit duration
status	string	No	Shipment status
created_at	timestamp	No	Shipment creation time
updated_at	timestamp	No	Last update

Privacy rule

Do not store unnecessary personal/customer PII.

4. Route Node / Checkpoint Data

File:

data/silver/checkpoints.parquet

This is the node layer of your graph.

Column	Type	Required	Description
node_id	string	Yes	Unique node ID
node_name	string	Yes	Node name
node_type	string	Yes	port, canal, customs, border, warehouse, terminal, hub, checkpoint
latitude	float	Yes	Latitude
longitude	float	Yes	Longitude
country_code	string	No	Country
region	string	No	State/province/region
city	string	No	City
timezone	string	No	Local timezone
capacity	float	No	Known capacity
capacity_unit	string	No	Capacity unit
operational_status	string	No	Normal/restricted/closed
opening_time	string	No	Operating hours
closing_time	string	No	Operating hours
source	string	Yes	Source
geometry	geometry	No	Point geometry

5. Route Edge / Transport Segment Data**File:**

data/silver/route_edges.parquet

Represents connections between checkpoints.

Column	Type	Required	Description
edge_id	string	Yes	Unique segment ID
route_id	string	Yes	Route identifier
source_node_id	string	Yes	Starting node
target_node_id	string	Yes	Ending node
transport_mode	string	Yes	Sea/road/rail/air
distance_km	float	Yes	Segment distance
planned_travel_hours	float	No	Planned duration
historical_travel_hours_mean	float	No	Historical mean
historical_travel_hours_std	float	No	Historical variability
historical_travel_hours_p50	float	No	P50
historical_travel_hours_p90	float	No	P90
historical_travel_hours_p95	float	No	P95
speed_limit	float	No	Applicable limit
route_type	string	No	Main/alternate/etc.
route_class	string	No	Route classification
risk_zone	string	No	Known risk zone
geometry	geometry	No	Segment geometry
direction	string	No	Direction
is_alternative	boolean	Yes	Alternative-route indicator
operational_status	string	No	Open/restricted/closed

6. AIS / Vessel Movement Data

File:

data/raw/ais/

data/silver/vessel_positions/

This is one of the most important datasets.

Raw AIS columns

Column	Type	Required
ais_record_id	string	Yes
vessel_id	string	Yes
mmsi	string	No
imo	string	No
timestamp	timestamp	Yes
latitude	float	Yes
longitude	float	Yes
speed_over_ground	float	No
course_over_ground	float	No
heading	float	No
navigation_status	string	No
vessel_type	string	No
vessel_name	string	No
length_m	float	No
width_m	float	No
draught_m	float	No
destination	string	No
eta	timestamp	No
call_sign	string	No
flag_country	string	No

7. Derived AIS Movement Features

File:

data/silver/vessel_movement_features.parquet

Column	Type	Description
vessel_id	string	Vessel
timestamp	timestamp	Observation time
latitude	float	Position
longitude	float	Position
speed_kmh	float	Converted speed
distance_since_previous_km	float	Movement distance
time_since_previous_min	float	Time difference
calculated_speed_kmh	float	Speed calculated from positions
speed_anomaly_flag	boolean	Impossible/unusual speed
position_anomaly_flag	boolean	Invalid position
stopped_flag	boolean	Vessel stopped
anchored_flag	boolean	Vessel anchored
moving_flag	boolean	Vessel moving
route_segment_id	string	Matched route edge
nearest_checkpoint_id	string	Nearest checkpoint
distance_to_checkpoint_km	float	Distance to checkpoint
time_in_segment_hours	float	Time spent in segment
checkpoint_arrival_time	timestamp	Derived arrival
checkpoint_departure_time	timestamp	Derived departure

8. Vessel / Traffic Aggregation Data

File:

data/features/traffic/

Aggregate AIS by checkpoint, port, route segment and time window.

Column	Type
location_id	string
node_id	string
timestamp	timestamp
date	date
vessel_count	integer
unique_vessel_count	integer
moving_vessel_count	integer
stopped_vessel_count	integer
anchored_vessel_count	integer
average_speed_kmh	float
median_speed_kmh	float
speed_std_kmh	float
vessel_density	float
traffic_volume	float
traffic_7d_avg	float
traffic_30d_avg	float
traffic_change_pct	float
stopped_vessel_change_pct	float
average_wait_hours	float
anchorage_pressure	float

9. Port Activity Data

File:

data/raw/ports/

data/silver/port_activity/

Do not assume every source provides every field.

Column	Type	Required
port_id	string	Yes
port_name	string	Yes
timestamp	timestamp	Yes
date	date	Yes
activity_type	string	Yes
arrivals	integer	No
departures	integer	No
port_calls	integer	No
vessel_count	integer	No
anchored_vessel_count	integer	No
berthed_vessel_count	integer	No
average_port_time_hours	float	No
median_port_time_hours	float	No
cargo_volume	float	No
cargo_volume_unit	string	No
container_volume	float	No
container_volume_unit	string	No
berth_occupancy	float	No
capacity	float	No

Column	Type	Required
operational_status	string	No
source	string	Yes

10. Port Activity Derived Features

Column	Type
port_id	string
timestamp	timestamp
activity_1d	float
activity_3d_avg	float
activity_7d_avg	float
activity_14d_avg	float
activity_30d_avg	float
activity_90d_avg	float
activity_change_1d_pct	float
activity_change_7d_pct	float
activity_change_30d_pct	float
activity_zscore	float
activity_anomaly	float
activity_anomaly_flag	boolean
congestion_index	float
congestion_level	string
port_pressure_index	float
operational_risk_score	float

11. Weather Location Data

File:

data/silver/weather_locations.parquet

Column	Type
location_id	string
location_name	string
location_type	string
node_id	string
latitude	float
longitude	float
country_code	string
timezone	string

This must remain **generic**.

Do not hard-code Frankfurt, Hamburg, Suez, India, etc.

12. Raw Weather Data

File:

data/raw/weather/

data/silver/weather/

Column	Type
location_id	string
timestamp	timestamp
latitude	float
longitude	float
temperature_2m	float

Column	Type
relative_humidity_2m	float
precipitation	float
rain	float
snowfall	float
weather_code	integer
surface_pressure	float
wind_speed_10m	float
wind_direction_10m	float
wind_gusts_10m	float
cloud_cover	float
visibility	float
source	string

Only request/store variables actually available from the selected weather provider.

13. Weather Derived Features

Column	Type
location_id	string
timestamp	timestamp
temperature_6h_avg	float
temperature_24h_avg	float
precipitation_3h_sum	float
precipitation_6h_sum	float
precipitation_24h_sum	float
wind_speed_3h_avg	float

Column	Type
wind_speed_6h_avg	float
wind_speed_24h_avg	float
wind_gust_3h_max	float
wind_gust_6h_max	float
wind_gust_24h_max	float
pressure_change_3h	float
pressure_change_6h	float
pressure_change_24h	float
weather_severity	float
weather_risk_level	string
extreme_weather_flag	boolean
weather_anomaly	float

14. Historical Checkpoint Outcome Data

This is the **ground-truth foundation for ML**.

File:

data/silver/checkpoint_outcomes.parquet

Column	Type	Required
observation_id	string	Yes
shipment_id	string	No
route_id	string	Yes
checkpoint_id	string	Yes
checkpoint_type	string	Yes
transport_mode	string	Yes

Column	Type	Required
planned_arrival_time	timestamp	No
actual_arrival_time	timestamp	Yes
planned_departure_time	timestamp	No
actual_departure_time	timestamp	No
planned_duration_hours	float	No
actual_duration_hours	float	No
delay_hours	float	Yes
delay_flag	integer	Yes
early_arrival_flag	boolean	No
data_timestamp	timestamp	Yes
source	string	Yes

Target definitions

delay_hours =

actual_arrival_time - planned_arrival_time

and:

delay_flag =

1 if delay_hours > configured_delay_threshold

0 otherwise

The threshold must be configurable.

Example:

delay_threshold_hours = 6

Do **not** define the target using weather.

15. Historical Delay Behaviour Features

Derived from actual historical outcomes.

File:

data/features/historical_delay/

Column	Type
checkpoint_id	string
as_of_timestamp	timestamp
historical_observation_count	integer
historical_delay_mean	float
historical_delay_median	float
historical_delay_std	float
historical_delay_min	float
historical_delay_max	float
historical_delay_p50	float
historical_delay_p75	float
historical_delay_p80	float
historical_delay_p90	float
historical_delay_p95	float
checkpoint_delay_rate	float
recent_delay_rate	float
delay_rate_7d	float
delay_rate_30d	float
delay_rate_90d	float
delay_rate_365d	float
delay_mean_7d	float
delay_mean_30d	float
delay_mean_90d	float

Column	Type
delay_mean_365d	float
delay_std_7d	float
delay_std_30d	float
delay_std_90d	float
delay_ewma	float
delay_trend	float
delay_volatility	float
seasonal_delay_mean	float

All rolling statistics must be calculated using **only information available before/as of the prediction timestamp**.

16. Temporal Features

These are mostly derived rather than externally collected.

Column	Type
timestamp	timestamp
hour	integer
minute	integer
day_of_week	integer
day_name	string
week_of_year	integer
month	integer
month_name	string
quarter	integer
day_of_year	integer

Column	Type
is_weekend	boolean
is_month_start	boolean
is_month_end	boolean
is_quarter_start	boolean
is_quarter_end	boolean
season	string
days_since_departure	float
hours_since_departure	float
days_to_required_delivery	float
hours_to_required_delivery	float
route_progress_pct	float
remaining_distance_km	float
distance_travelled_km	float

17. Holiday / Calendar Data

File:

data/silver/holidays.parquet

Column	Type
holiday_id	string
date	date
country_code	string
region	string
holiday_name	string
holiday_type	string

Column	Type
is_public_holiday	boolean
is_operational_holiday	boolean
holiday_duration_days	integer
Derived:	
is_holiday	
days_to_holiday	
days_since_holiday	
holiday_period_flag	

18. External Disruption / Event Data

File:

data/raw/events/

data/silver/disruptions/

Column	Type	Required
event_id	string	Yes
event_timestamp	timestamp	Yes
event_type	string	Yes
event_subtype	string	No
event_title	string	No
description	string	No
location_name	string	No
latitude	float	No
longitude	float	No
country_code	string	No

Column	Type	Required
region	string	No
affected_node_id	string	No
affected_route_id	string	No
affected_edge_id	string	No
start_time	timestamp	No
end_time	timestamp	No
severity	float	No
severity_level	string	No
event_status	string	No
source	string	Yes
source_url	string	No
confidence	float	No
verified_flag	boolean	No

Event types

The system should support categories such as:

STRIKE

PORT_CLOSURE

PORT_CONGESTION

CANAL_BLOCKAGE

CANAL_RESTRICTION

INFRASTRUCTURE_FAILURE

SEVERE_WEATHER

FLOOD

STORM

FIRE

EARTHQUAKE

BORDER_CLOSURE

CUSTOMS_DISRUPTION

GEOPOLITICAL_EVENT

CONFLICT

SECURITY_EVENT

TRANSPORT_DISRUPTION

OTHER

Do not assume every category exists in every source.

19. Event-Derived Features

Column	Type
event_presence	boolean
event_count_nearby	integer
event_severity	float
event_recency_hours	float
event_distance_km	float
nearest_event_distance_km	float
active_event_count	integer
disruption_flag	boolean
major_disruption_flag	boolean
geopolitical_risk	float
infrastructure_risk	float
operational_disruption_risk	float
event_exposure_score	float
route_disruption_exposure	float

20. News / NLP Event Data

This is the optional AI/NLP layer.

File:

data/raw/news/

data/silver/nlp_events/

Column	Type
document_id	string
source	string
source_url	string
published_at	timestamp
retrieved_at	timestamp
headline	string
article_text	string
language	string
location_name	string
latitude	float
longitude	float
country_code	string
event_type	string
event_subtype	string
event_start_time	timestamp
event_end_time	timestamp
severity	float
severity_level	string

Column	Type
affected_asset	string
affected_port_id	string
affected_route_id	string
affected_node_id	string
event_presence	boolean
event_recency_hours	float
event_distance_km	float
extraction_confidence	float
verification_status	string

Important

The LLM should produce **structured event intelligence**.

It should not directly produce:

ETA = 17 hours late

ETA remains the responsibility of the predictive pipeline.

21. Data Quality / Anomaly Data

File:

data/silver/data_quality_flags.parquet

Column	Type
record_id	string
dataset_name	string
timestamp	timestamp
quality_status	string
missing_value_flag	boolean

Column	Type
duplicate_flag	boolean
invalid_coordinate_flag	boolean
impossible_speed_flag	boolean
timestamp_order_error_flag	boolean
future_timestamp_flag	boolean
invalid_range_flag	boolean
inconsistent_record_flag	boolean
outlier_flag	boolean
quality_score	float
quality_flags	string/list
correction_applied	boolean
correction_method	string

22. Regime / Change-Point Data

File:

data/features/regimes.parquet

Column	Type
location_id	string
checkpoint_id	string
route_id	string
timestamp	timestamp
regime_state	string
regime_id	string
change_point_flag	boolean

Column	Type
change_point_score	float
regime_duration_hours	float
regime_duration_days	float
activity_anomaly	float
delay_anomaly	float
congestion_anomaly	float
weather_anomaly	float
disruption_anomaly	float
regime_confidence	float

Recommended states:

NORMAL

DETERIORATING

DISRUPTION

SEVERE_DISRUPTION

RECOVERY

The exact regime algorithm can be implemented using change-point detection such as ruptures.

23. Checkpoint Context Feature Table

This is where the different datasets begin to merge.

File:

data/features/checkpoint_features.parquet

Column	Type
observation_id	string
shipment_id	string

Column	Type
route_id	string
checkpoint_id	string
checkpoint_type	string
timestamp	timestamp
latitude	float
longitude	float
transport_mode	string
distance_to_next_km	float
remaining_distance_km	float
route_progress_pct	float
planned_travel_hours	float
historical_delay_mean	float
historical_delay_std	float
historical_delay_p90	float
historical_delay_p95	float
checkpoint_delay_rate	float
recent_delay_rate	float
delay_mean_7d	float
delay_mean_30d	float
delay_mean_90d	float
delay_mean_365d	float
delay_ewma	float
delay_trend	float

Column	Type
vessel_count	float
vessel_density	float
average_speed_kmh	float
stopped_vessel_count	float
anchored_vessel_count	float
activity_7d_avg	float
activity_change_pct	float
activity_anomaly	float
congestion_index	float
temperature_2m	float
relative_humidity_2m	float
precipitation	float
wind_speed_10m	float
wind_gusts_10m	float
surface_pressure	float
weather_code	float
wind_speed_6h_avg	float
precipitation_6h_sum	float
wind_gust_6h_max	float
weather_severity	float
weather_risk_level	string
hour	integer
day_of_week	integer

Column	Type
week_of_year	integer
month	integer
day_of_year	integer
is_weekend	boolean
is_holiday	boolean
season	string
days_since_departure	float
days_to_required_delivery	float
regime_state	string
change_point_flag	boolean
regime_duration_hours	float
disruption_flag	boolean
event_presence	boolean
event_severity	float
event_recency_hours	float
event_distance_km	float
geopolitical_risk	float
event_exposure_score	float

24. ML Training Dataset ★★ ★

File:

data/ml/ml_training_data.parquet

This is the dataset given to LightGBM.

Identifiers

observation_id

shipment_id

route_id

checkpoint_id

checkpoint_type

timestamp

Route features

transport_mode

distance_to_next_km

remaining_distance_km

route_progress_pct

planned_travel_hours

Historical features

historical_delay_mean

historical_delay_std

historical_delay_p50

historical_delay_p90

historical_delay_p95

checkpoint_delay_rate

recent_delay_rate

delay_mean_7d

delay_mean_30d

delay_mean_90d

delay_mean_365d

delay_std_7d

delay_std_30d

delay_std_90d

delay_ewma

delay_trend

delay_volatility

seasonal_delay_mean

Port / activity features

vessel_count

unique_vessel_count

moving_vessel_count

stopped_vessel_count

anchored_vessel_count

average_speed_kmh

median_speed_kmh

vessel_density

traffic_volume

traffic_7d_avg

traffic_30d_avg

traffic_change_pct

stopped_vessel_change_pct

average_wait_hours

anchorage_pressure

activity_7d_avg

activity_14d_avg

activity_30d_avg

activity_90d_avg

activity_change_1d_pct

activity_change_7d_pct

activity_change_30d_pct

activity_anomaly

congestion_index

port_pressure_index
operational_risk_score

Weather features

temperature_2m
relative_humidity_2m
precipitation
rain
snowfall
weather_code
surface_pressure
wind_speed_10m
wind_direction_10m
wind_gusts_10m
cloud_cover
visibility
temperature_6h_avg
temperature_24h_avg
precipitation_3h_sum
precipitation_6h_sum
precipitation_24h_sum
wind_speed_3h_avg
wind_speed_6h_avg
wind_speed_24h_avg
wind_gust_3h_max
wind_gust_6h_max
wind_gust_24h_max
pressure_change_3h
pressure_change_6h

pressure_change_24h

weather_severity

weather_risk_level

extreme_weather_flag

weather_anomaly

Temporal features

hour

minute

day_of_week

week_of_year

month

quarter

day_of_year

is_weekend

is_month_start

is_month_end

is_quarter_start

is_quarter_end

season

is_holiday

days_since_departure

hours_since_departure

days_to_required_delivery

hours_to_required_delivery

Regime features

regime_state

change_point_flag

change_point_score

regime_duration_hours

regime_duration_days

activity_anomaly

delay_anomaly

congestion_anomaly

weather_anomaly

regime_confidence

Disruption features

disruption_flag

major_disruption_flag

event_presence

event_count_nearby

event_severity

event_recency_hours

event_distance_km

nearest_event_distance_km

active_event_count

geopolitical_risk

infrastructure_risk

operational_disruption_risk

event_exposure_score

route_disruption_exposure

Targets

delay_flag

delay_hours

These are the **only outcome/target columns**.

25. ML Prediction Output

File:

data/ml/predictions.parquet

Column	Type
observation_id	string
shipment_id	string
route_id	string
checkpoint_id	string
prediction_timestamp	timestamp
model_version	string
delay_probability	float
predicted_delay_flag	integer
expected_delay_hours	float
delay_p50_hours	float
delay_p80_hours	float
delay_p90_hours	float
delay_p95_hours	float
risk_score	float
risk_level	string
prediction_confidence	float

26. Checkpoint Risk Data**File:**

data/outputs/checkpoint_risk.parquet

Column	Type
shipment_id	string

Column	Type
route_id	string
checkpoint_id	string
checkpoint_type	string
risk_score	float
risk_level	string
delay_probability	float
expected_delay_hours	float
delay_p50_hours	float
delay_p80_hours	float
delay_p90_hours	float
delay_p95_hours	float
uncertainty_hours	float
regime_state	string
disruption_flag	boolean
weather_risk_level	string
congestion_index	float
criticality_score	float
downstream_impact_score	float
deadline_risk_contribution	float
prediction_timestamp	timestamp

27. Delay Propagation Data

File:

data/outputs/delay_propagation.parquet

Column	Type
shipment_id	string
route_id	string
source_checkpoint_id	string
target_checkpoint_id	string
edge_id	string
propagated_delay_hours	float
propagated_delay_p50	float
propagated_delay_p80	float
propagated_delay_p90	float
propagated_delay_p95	float
propagation_probability	float
cumulative_delay_hours	float
downstream_impact_score	float
timestamp	timestamp

28. Monte Carlo Simulation Output

File:

data/outputs/monte_carlo_results.parquet

For 10,000+ simulations:

Column	Type
simulation_id	integer
shipment_id	string
route_id	string
simulation_timestamp	timestamp

Column	Type
simulated_arrival_time	timestamp
simulated_delay_hours	float
simulated_transit_hours	float
deadline_missed	boolean
total_propagated_delay_hours	float
critical_checkpoint_id	string
scenario_id	string

29. ETA Distribution Output

File:

data/outputs/eta_distribution.parquet

Column	Type
shipment_id	string
route_id	string
calculation_timestamp	timestamp
planned_arrival_time	timestamp
eta_p50	timestamp
eta_p80	timestamp
eta_p90	timestamp
eta_p95	timestamp
delay_p50_hours	float
delay_p80_hours	float
delay_p90_hours	float
delay_p95_hours	float

Column	Type
expected_delay_hours	float
eta_mean	timestamp
eta_std_hours	float
deadline	timestamp
deadline_miss_probability	float
on_time_probability	float

30. Critical Checkpoint Attribution

File:

data/outputs/critical_checkpoints.parquet

Column	Type
shipment_id	string
route_id	string
checkpoint_id	string
checkpoint_rank	integer
risk_score	float
expected_delay_contribution_hours	float
uncertainty_contribution_hours	float
deadline_risk_contribution	float
downstream_impact_score	float
criticality_score	float
criticality_reason	string
calculation_timestamp	timestamp

31. SHAP / Explainability Output

File:

data/outputs/shap_explanations.parquet

Column	Type
observation_id	string
shipment_id	string
checkpoint_id	string
prediction_timestamp	timestamp
feature_name	string
feature_value	float/string
shap_value	float
contribution_direction	string
feature_rank	integer
model_output	float
model_version	string

Example:

congestion_index	+0.31	↑ risk
historical_delay_p90	+0.22	↑ risk
weather_severity	+0.15	↑ risk
recent_delay_rate	+0.11	↑ risk

32. What-If / Counterfactual Scenario Input

File:

data/scenarios/what_if_scenarios.parquet

Column	Type
scenario_id	string

Column	Type
shipment_id	string
base_route_id	string
created_at	timestamp
scenario_name	string
scenario_type	string
target_node_id	string
target_edge_id	string
target_feature	string
baseline_value	float
intervention_value	float
change_pct	float
duration_hours	float
start_time	timestamp
end_time	timestamp
description	string

Example:

scenario_type = CONGESTION_INCREASE

target_node_id = X

target_feature = congestion_index

baseline_value = 0.42

intervention_value = 0.75

change_pct = 78.6

33. What-If Scenario Output

File:

data/outputs/scenario_results.parquet

Column	Type
scenario_id	string
shipment_id	string
base_route_id	string
scenario_route_id	string
baseline_eta_p50	timestamp
scenario_eta_p50	timestamp
baseline_eta_p90	timestamp
scenario_eta_p90	timestamp
baseline_deadline_miss_probability	float
scenario_deadline_miss_probability	float
baseline_expected_delay_hours	float
scenario_expected_delay_hours	float
eta_change_hours	float
risk_change_pct	float
uncertainty_change_hours	float
recommendation	string

34. Causal / Intervention Data

For the advanced intervention-aware component.

File:

data/outputs/intervention_analysis.parquet

Column	Type
intervention_id	string

Column	Type
shipment_id	string
base_route_id	string
intervention_type	string
target_node_id	string
target_edge_id	string
baseline_risk	float
intervention_risk	float
risk_delta	float
baseline_expected_delay_hours	float
intervention_expected_delay_hours	float
delay_reduction_hours	float
baseline_eta_p90	timestamp
intervention_eta_p90	timestamp
eta_improvement_hours	float
baseline_deadline_miss_probability	float
intervention_deadline_miss_probability	float
deadline_risk_reduction	float
uncertainty_reduction_hours	float
intervention_score	float
recommendation	string

35. Alternative Route Comparison

File:

data/outputs/route_comparison.parquet

Column	Type
comparison_id	string
shipment_id	string
route_id	string
route_rank	integer
route_name	string
distance_km	float
planned_transit_hours	float
expected_transit_hours	float
expected_delay_hours	float
delay_probability	float
eta_p50	timestamp
eta_p80	timestamp
eta_p90	timestamp
eta_p95	timestamp
deadline_miss_probability	float
disruption_exposure	float
geopolitical_risk	float
weather_risk	float
congestion_risk	float
uncertainty_hours	float
critical_checkpoint_count	integer
route_risk_score	float
risk_adjusted_score	float

Column	Type
recommended_flag	boolean
recommendation_reason	string

36. Dynamic Risk State

For the live/current dashboard.

File:

data/outputs/current_risk_state.parquet

Column	Type
shipment_id	string
route_id	string
checkpoint_id	string
last_updated_at	timestamp
current_latitude	float
current_longitude	float
current_route_segment_id	string
current_speed_kmh	float
current_regime_state	string
current_congestion_index	float
current_weather_risk	float
current_disruption_risk	float
current_delay_probability	float
current_expected_delay_hours	float
current_eta_p50	timestamp
current_eta_p90	timestamp

Column	Type
current_deadline_miss_probability	float
risk_change_from_previous	float
risk_trend	string
recalculation_reason	string
model_version	string

37. Model Evaluation Data

File:

data/ml/model_evaluation.parquet

Classification

model_version

dataset_split

accuracy

precision

recall

f1_score

roc_auc

pr_auc

log_loss

brier_score

calibration_error

Regression

mae

rmse

r2

median_absolute_error

Quantile

quantile

pinball_loss

coverage

calibration_error

Metadata

train_start_time

train_end_time

validation_start_time

validation_end_time

test_start_time

test_end_time

feature_count

training_row_count

model_version

training_timestamp

38. Graph Data Model

Your graph should contain at least:

Node properties

node_id

node_name

node_type

latitude

longitude

country_code

region

operational_status

capacity
current_risk_score
delay_probability
expected_delay_hours
delay_p50
delay_p90
delay_p95
regime_state
disruption_flag
weather_risk
congestion_index
criticality_score

Edge properties

edge_id
source_node_id
target_node_id
route_id
transport_mode
distance_km
planned_travel_hours
historical_travel_hours_mean
historical_travel_hours_std
travel_time_p50
travel_time_p90
travel_time_p95
current_risk
disruption_exposure
weather_exposure

congestion_exposure

propagated_delay_hours

39. Recommended Final Parquet Structure

data/

|

| └─ raw/

|

| └─ ais/

|

| └─ year=YYYY/month=MM/*.parquet

|

|

| └─ weather/

|

| └─ year=YYYY/month=MM/*.parquet

|

|

| └─ ports/

|

| └─ year=YYYY/*.parquet

|

|

| └─ events/

|

| └─ year=YYYY/*.parquet

|

|

| └─ news/

|

| └─ year=YYYY/month=MM/*.parquet

|

|

| └─ shipments/

|

| └─ *.parquet

|

|

| └─ silver/

|

| └─ shipments.parquet

- | |— checkpoints.parquet
- | |— route_edges.parquet
- | |— vessel_positions/
- | |— vessel_movement_features/
- | |— port_activity/
- | |— weather/
- | |— disruptions/
- | |— holidays.parquet
- | |— checkpoint_outcomes.parquet
- | |— data_quality_flags.parquet
- |
- |— features/
- | |— traffic/
- | |— port_features/
- | |— weather_features/
- | |— historical_delay/
- | |— temporal_features/
- | |— disruption_features/
- | |— regimes.parquet
- | |— checkpoint_features.parquet
- |
- |— ml/
- | |— ml_training_data.parquet
- | |— train.parquet
- | |— validation.parquet
- | |— test.parquet
- | |— predictions.parquet

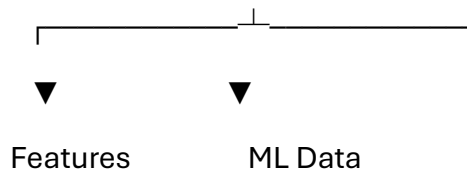
```
| └─ model_evaluation.parquet
|
|─ scenarios/
|   └─ what_if_scenarios.parquet
|
└─ outputs/
    ├── checkpoint_risk.parquet
    ├── delay_propagation.parquet
    ├── monte_carlo_results.parquet
    ├── eta_distribution.parquet
    ├── critical_checkpoints.parquet
    ├── shap_explanations.parquet
    ├── scenario_results.parquet
    ├── intervention_analysis.parquet
    ├── route_comparison.parquet
    └─ current_risk_state.parquet
```

40. DuckDB Architecture

Do **not** duplicate every Parquet dataset unnecessarily inside DuckDB.

Use:

```
PARQUET
|
▼
DuckDB
|
SQL / joins / aggregation
|
```



Recommended DuckDB views:

ais

weather

port_activity

checkpoints

route_edges

checkpoint_outcomes

disruptions

holidays

checkpoint_features

ml_training_data

predictions

checkpoint_risk

eta_distribution

route_comparison

For example:

```
CREATE VIEW weather AS
```

```
SELECT *
```

```
FROM read_parquet('data/silver/weather/**/*.*parquet');
```

and:

```
CREATE VIEW checkpoint_features AS
```

```
SELECT *
```

```
FROM read_parquet('data/features/checkpoint_features.parquet');
```

Therefore:

Parquet = persistent storage

DuckDB = analytical/query engine

Neo4j = graph database

NetworkX = in-memory graph processing

You don't need to force every piece of data into every system.

41. Data Ownership by Technology

Data	Primary storage/technology
Raw AIS	Parquet
Clean AIS	Parquet
Weather	Parquet
Port activity	Parquet
Historical outcomes	Parquet
Events	Parquet
News/NLP events	Parquet
Feature tables	Parquet
SQL analytics	DuckDB
ML training	Parquet + DuckDB
Model artifacts	.joblib / .txt / native model format
Route graph	Neo4j + NetworkX
GNN graph tensors	PyTorch/PyG objects
Monte Carlo results	Parquet
Dashboard/API	Reads processed outputs

42. Critical Rules for OpenCode

The implementation should follow these rules exactly.

1. NEVER fabricate missing real-world data.

2. NEVER fabricate delay labels.
3. delay_hours must originate from observed operational outcomes.
4. delay_flag must be derived from actual delay_hours.
5. Weather is a predictor, NOT the delay label.
6. Port activity is a predictor, NOT the delay label.
7. AIS is a predictor/source of operational outcomes, NOT a fabricated label.
8. Every time-dependent feature must respect prediction time.
9. No future information may enter historical training features.
10. Preserve source timestamps.
11. Preserve source provenance.
12. Keep raw and processed data separate.
13. Store persistent datasets as Parquet.
14. Use DuckDB for analytical querying and feature joins.
15. Use GeoParquet where appropriate for spatial datasets.

16. Do not hard-code Frankfurt → India.
17. Do not hard-code specific ports, canals or shipment IDs.
18. Route/checkpoint configuration must be data-driven.
19. All risk calculations must work for arbitrary valid routes.
20. Monte Carlo simulations may be synthetic scenarios,
but their probability distributions must come from real historical data.
21. What-if scenarios modify features/conditions,
but must not overwrite the underlying real data.
22. Real disruptions must not automatically be deleted as statistical outliers.
23. Data anomalies and real-world disruptions must be treated separately.
24. Model versions must be recorded with predictions.
25. Every prediction should have a prediction timestamp.
26. Every rolling historical feature must have an as-of timestamp.
27. Missing source fields should remain NULL/NaN,
not be filled with invented values.

28. Feature availability should be tracked.

29. Train/validation/test splitting must be chronological.

30. Final model evaluation must use future-held-out data.

43. Minimum Required Data for the Actual 2-Day Prototype

Although the schema above is the **complete target architecture**, don't try to collect every possible column before your prototype works.

The absolute core is:

1. Historical movement/AIS data
2. Actual checkpoint arrival/departure or enough movement data to derive them
3. Route/checkpoint coordinates
4. Open-Meteo historical weather
5. Port/vessel activity
6. Historical delay statistics
7. Temporal features
8. Real disruption information

From these, you can build:

Real Data

↓

Data Quality

↓

Historical Delay

↓

Weather + Activity + Temporal + Disruption Features

↓

LightGBM

↓

Checkpoint Delay Probability

↓

Delay Magnitude

↓

P50/P80/P90/P95

↓

Graph Propagation

↓

10,000 Monte Carlo Simulations

↓

Probabilistic ETA

↓

Deadline-Miss Probability

↓

Critical Checkpoint

↓

SHAP

↓

What-If

↓

Route Comparison

That is the **minimum complete end-to-end LOGIX pipeline**.

The larger schema above should be treated as the **canonical data contract**, while the implementation can initially populate only the fields genuinely available from the real datasets you select.